Gopal Kataria

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Education

International Institute of Information Technology, Hyderabad (IIIT-H) — *CGPA: 7.79/10*
B.Tech + MS (By Research), Electronics and Communication Engineering — *July 2023 – May 2028 (Expected)*

JEE Mains 2023 — *99.17 Percentile*
Vasantrao Dempo Higher Secondary School, Cujira, Goa — *March 2023*

Experience

AI Research Intern, ixigo — *May 2026 – Jul 2026*
Audio Preprocessing, Pandas, Librosa, PyTorch, PEFT/LoRA, Hugging Face Transformers, vLLM, Modal

- Engineered a Hindi/Hinglish TTS corpus from real call-center recordings, creating targeted datasets for numerals, travel terms, Hinglish, and Indian place names to improve pronunciation.
- LoRA fine-tuned and served an open-source Hindi/Hinglish TTS model with vLLM, achieving ~140 ms streaming first-byte latency comparable to the production commercial provider.

Systems Administrator & Web Developer, DFL, IIIT-H — *Jun 2025 – Dec 2025*
Linux, Docker, Nginx, CI/CD, SvelteKit, PostgreSQL

- Owned the end-to-end deployment of a production student portal (SvelteKit, PostgreSQL), provisioning Linux infrastructure with Docker, Nginx, and CI/CD while supporting 200+ students with course registration and online payments.

Undergraduate Researcher, SPCRC, IIIT-H — *May 2025 – Present*
Statistical Detection Theory, Sequential Analysis, Change-Point Detection, Time Series Analysis, Copula Methods, MATLAB/Octave

- Developed a novel change-point detection framework for transmitted wireless signals with applications in surveillance and defense communications, deriving theoretical performance guarantees and backing them with numerical and simulation-based analysis; manuscript in preparation.

Projects

LANdrop — GitHub
C, Linux Systems Programming, Socket Programming, OpenSSL/TLS, epoll
Wrote a C-based P2P LAN file-transfer daemon featuring UDP peer discovery, an epoll-driven event loop, OpenSSL/TLS-secured chunked transfers with Trust-On-First-Use pinning, SHA-256 integrity verification, and a persistent trust/configuration store.

Event Registration Infrastructure — Live Link
Go, PostgreSQL, React, Azure Communication Services, Nginx
Built and scaled IIIT-H's festival ticketing system (Go Fiber, Postgres, React) to 11,000+ registrations and 6,000+ check-ins, absorbing a 1,500% traffic spike and clearing a 4,000-email backlog overnight, keeping new registrations emails delivering under 20s.

Distributed Workflow Execution Platform — GitHub
Python, Distributed Systems, FastAPI, Kafka, Docker, MongoDB
Built an agentic AI workflow orchestration platform with a FastAPI control plane and isolated Docker workers, using Kafka for asynchronous artifact exchange and an Inspect → Build → Deploy pipeline for automated containerized service deployment..

Knowledge Unlearning in LLMs using SAEs — GitHub
PyTorch, Transformers, Sparse Autoencoders, Mechanistic Interpretability, LLMs
Used sparse autoencoders (16,384-feature dictionary) via TransformerLens on llama-7B's residual stream to ablate Harry Potter specific features, cutting recall by 77% with only +0.1% perplexity cost and 4x tighter localization.

Technical Skills

Languages: Python, C/C++, Go, TypeScript, SQL, Bash | **Backend & APIs:** FastAPI, Fiber, SvelteKit, React, REST APIs, WebSockets | **Infrastructure & Cloud:** Linux, Docker, Kafka, PostgreSQL, MongoDB, Nginx, Azure, Git, CI/CD (GitHub Actions) | **Systems:** Distributed Systems, Concurrency, Event-Driven Architecture (epoll), Networking, TLS/Applied Cryptography | **Coursework:** Operating Systems, Database Systems, Computer Networks, DSA

Achievements

Qualified for the online semifinals (**Top 1,500** coders nationwide) in **Google Big Code**.
Qualified for the online semifinals (**Top 1,600** teams) in **Flipkart GRiD 8.0**.
Reached the **ANRF AISEHACK (IBM Theme)** semifinals (**Top 100** nationwide); placed **18th** in pre-finals and **20th** overall.
Ranked **337/16,000+** participants in the **Goldman Sachs India Hackathon (CS Arena)**.

Leadership & Roles

Coordinator, Open Source Developers Group (OSDG), IIIT-H — *Mar 2025 – Mar 2026*
Led a 25-member open-source community, running Bash/Git/Docker workshops for 100+ freshers and drawing 300+ hackathon participants, hosting events in collaboration with Jane Street, FOSS United, Coding Ninjas and Hyderabad Python User Group.

Tech Head, Felicity 2026, IIIT Hyderabad — *Jan 2026 – Mar 2026*
Led the engineering team behind Felicity 2026's registration platform, supporting 11,000+ registrations and 100+ paying teams with zero downtime through last-minute requirement changes.